

For example: If an attribute is a struct data type, that is readable and writable, then all fields of the struct are readable and writable.

New or optional data fields might not be recognized by a receiver, such as a legacy receiver. The data field description SHALL define default behavior (such as absence of behavior) when a new or optional data field is not present. It is recommended to define or use a feature when adding new or optional data fields, to better indicate conformance. It is recommended to define a [fallback value](#), such as a NULL value, that indicates such default behavior.

7.18.1. Nullable

When a data field value is required to designate an unknown, invalid, or undefined data value, and there is no obvious data value (e.g. zero), that is within the valid range to indicate this, the data field MAY be designated as nullable, so that an implemented instance of data MAY have the value of NULL.

In this context, these wordings have the same meaning:

- The data field has the value of null.
- The data field has the null value.
- The data field is the null value.
- The data field is null.
- The data field has the value of NULL.
- The data field has the NULL value.
- The data field is the NULL value.
- The data field is NULL.

Representation of NULL for the implementation of the Data Model is a consideration of the underlying encoding specification. The encoding layer SHALL have the capability to indicate NULL for any nullable data field. How the encoding layer indicates NULL is outside the scope of the Data Model specification.

All data fields MAY be defined to be nullable, regardless of data type.

A cluster specification SHALL define whether a data field is nullable. A cluster specification SHALL define the meaning of the NULL value.

Composite data types that have a length (i.e. octet string and list), and derived types that have those as the base type, SHALL NOT differentiate semantically between the NULL value and the empty (zero length) value. In particular a zero-length value SHALL be allowed for nullable values of these types no matter what other length constraints are imposed on the value, and SHALL have the same semantics as the NULL value.

A fabric-scoped list SHALL NOT be nullable.